

# Touchdot S3 Development Board Product Brief

Compact ESP32-S3 mini microcontroller development board inspired by Lilypad. Ideal for IoT, control systems, and creative electronics projects.

Version: 1.0

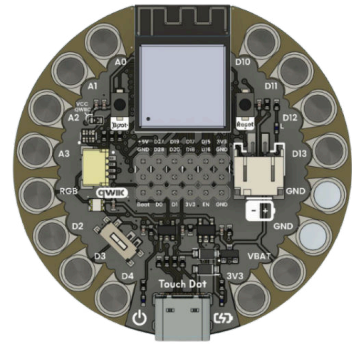
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## Introduction

The Unit Touchdot S3 is a compact and powerful development board based on the ESP32-S3 Mini, tailored for applications in wearable electronics, IoT devices, educational platforms, and smart automation. Inspired by the Lilypad design philosophy, it offers a low-profile, sewable form factor ideal for integration into textiles and compact enclosures, while delivering advanced wireless and processing capabilities.

Equipped with 2.4 GHz Wi-Fi and Bluetooth 5.0, the Touchdot S3 enables seamless communication with mobile devices, web services, and wireless sensor networks. Its modern interface includes a USB-C connector for programming and power delivery, onboard LiPo battery charging, and a standard QWIIC I<sup>2</sup>C port for rapid peripheral integration.

Designed for rapid prototyping and creative development, the Touchdot S3 bridges the gap between wearable design and embedded computing, offering a robust platform for students, makers, and engineers alike.



## Functional Description

- Integrated ESP32-S3 module with 2.4 GHz Wi-Fi and Bluetooth 5.0
- USB-C connector for power and programming
- 3.3V power rail compatible with low-voltage peripherals
- Built-in QWIIC connector for easy I<sup>2</sup>C module integration
- Micro SD card slot using SPI interface
- Onboard NeoPixel (WS2812) RGB LED

## Electrical Characteristics

- Operating voltage: 3.3V
- Max current draw: 500mA (with Wi-Fi active)
- GPIO logic level: 3.3V
- ADC resolution: 12-bit (0–4095)
- Touchpad sensitivity: configurable

## Applications

- Wearable electronics
- IoT sensor nodes
- Smart home and automation
- Educational tools for STEM
- Environmental monitoring
- Creative electronics and art-tech installations
- Smart Home
- Industrial Automation
- Health Care
- Consumer Electronics
- Smart Agriculture
- POS Machines

## Settings

### Interface Overview

| Interface | Signals / Pins      | Typical Use                             |
|-----------|---------------------|-----------------------------------------|
| UART      | Tx, Rx              | Serial terminal or sensor communication |
| I2C       | SDA, SCL            | QWIIC modules, OLED displays            |
| SPI       | MOSI, MISO, SCK, CS | External flash, TFT display             |
| USB       | D+, D-              | Native USB device or host               |

### Supported Pins

| Symbol | I/O           | Description               |
|--------|---------------|---------------------------|
| VCC    | Input         | Power supply (3.3V or 5V) |
| GND    | GND           | Ground connection         |
| IO     | Bidirectional | General-purpose I/O pins  |

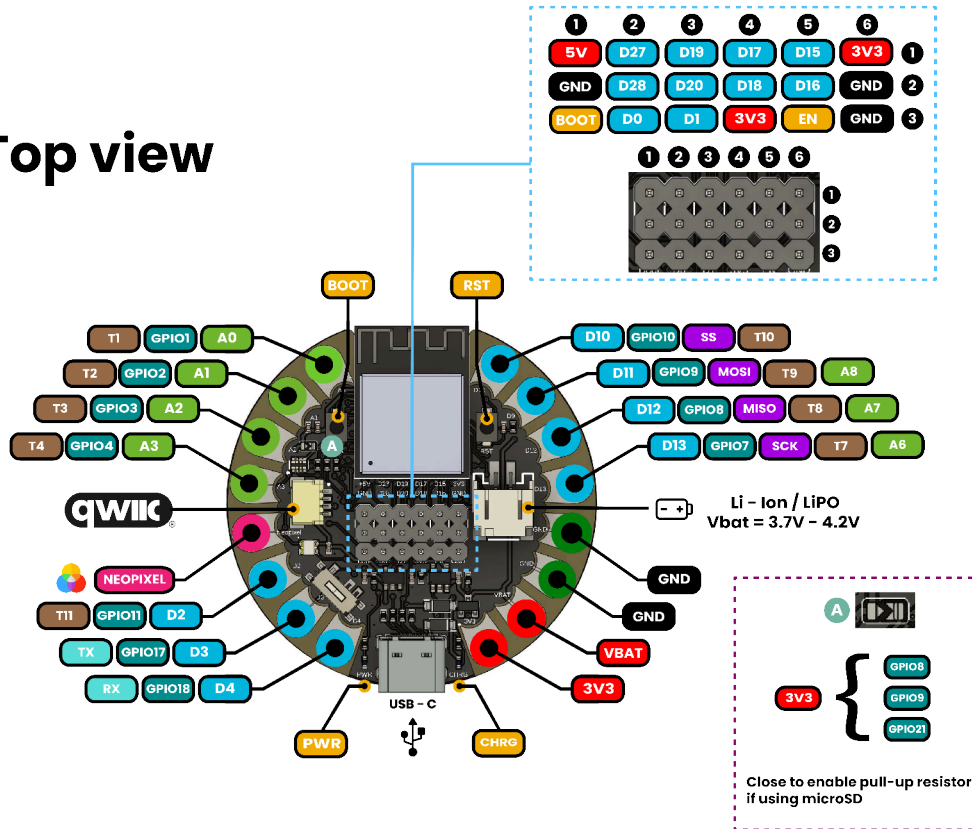
### Pin & Connector Layout

| Group    | Available pins | Suggested use                          |
|----------|----------------|----------------------------------------|
| GPIO     | D2 to D13      | Sensors, actuators                     |
| UART     | Tx and Rx      | Serial communication                   |
| TouchPad | T1 to T11      | Capacitive sensors for touch detection |
| Analog   | A0 to A8       | 12-bit (0–4095) resolution             |
| SPI      | Optional       | Displays, additional memory            |

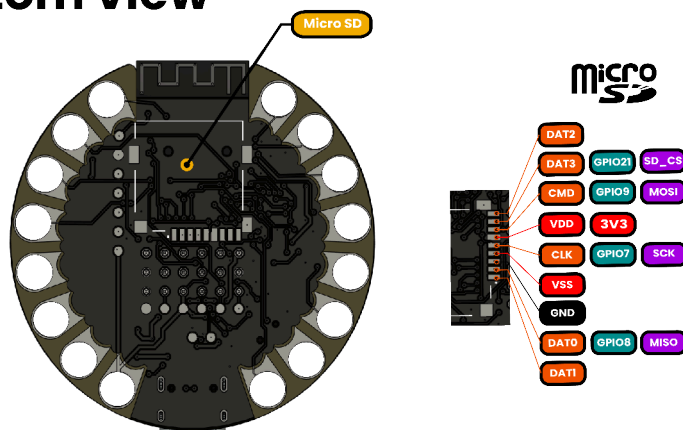
## Block Diagram

# UNIT Touchdot S3

## Top view



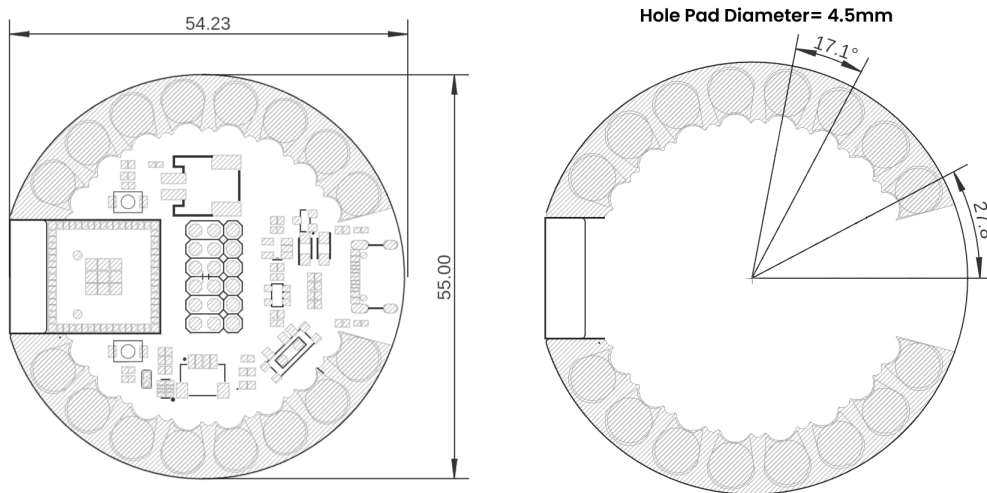
## Bottom view



## Description:

|                                                   |                                                                  |                                                |                                                     |
|---------------------------------------------------|------------------------------------------------------------------|------------------------------------------------|-----------------------------------------------------|
| <span style="color: red;">■</span> Supply voltage | <span style="color: blue;">■</span> GPIO compatible with Arduino | <span style="color: brown;">■</span> Touch Pad | <span style="color: orange;">■</span> Micro SD Data |
| <span style="color: black;">■</span> GND          | <span style="color: teal;">■</span> GPIO ESP32                   | <span style="color: green;">■</span> Analogic  | <span style="color: purple;">■</span> RGB           |
| <span style="color: yellow;">■</span> Components  | <span style="color: lightblue;">■</span> UART                    | <span style="color: purple;">■</span> SPI      | <span style="color: green;">■</span> Exposed pad    |

## Dimensions



Mechanical dimensions in millimeters of Touch Dot S3

## Usage

- Arduino IDE (ESP32 board manager)
- PlatformIO (ESP32-S3 support)
- ESP-IDF toolchain
- MicroPython firmware
- CircuitPython (via UF2 bootloader)

## Downloads

- Schematic PDF
- Board Dimensions DXF
- Pinout Diagram PNG

## Purchase

- Buy from UNIT Electronics
- Open product page